X

(https://swayam.gov.in) (https://swayam.gov.in/nc_details/NPTEL)

riya.soni17oct@gmail.com

NPTEL (https://swayam.gov.in/explorer?ncCode=NPTEL) » Introduction to Graph Algorithms (course)

**If already registered, click to check your payment status**


Course outline

**About NPTEL ()**

**How does an NPTEL online course work? ()**

**Week 1 Introduction and Principles of Algorithms ()**

**Week 2 Shortest Path Algorithms and problems ()**

**Week 3 Bellman Equations ()**


# Week 6 Assignment 6

**The due date for submitting this assignment has passed.**

**Due on 2025-09-03, 23:59 IST.**

## Assignment submitted on 2025-09-02, 13:13 IST

1) State True or False *1 point*

If $G$ has $m \geq n$ edges and unique heaviest edge $e$, then $e$ is NOT a part of any MST.

- True
- False

Yes, the answer is correct.
Score: 1

Accepted Answers:
*False*

2) State True or False *1 point*

Prim's algorithm does not work if $G$ has negative edges.

- True
- False

Yes, the answer is correct.
Score: 1

Accepted Answers:
*False*

3) State True or False *1 point*

Kruskal's algorithm will output an MST even if $G$ has a negative cycle.

- True
- False

**Week 4 Bellman Ford and Dijkstra Algorithm ()**

**Week 5 All Pair Shortest Path ()**

**Week 6 Prims and Kruskal's Algorithm ()**

Prims Algorithm (unit?unit=23&lesson=73)

Prims Algorithm P2 (unit?unit=23&lesson=74)

Kruskal's Algorithm (unit?unit=23&lesson=75)

Kruskal's Algorithm 2 (unit?unit=23&lesson=76)

Kruskal's Algorithm 3 (unit?unit=23&lesson=77)

Week 6 Lecture Material (unit?unit=23&lesson=67)

**Quiz: Week 6 Assignment 6 (assessment?name=102)**

Feedback Form (unit?

Yes, the answer is correct.
Score: 1

Accepted Answers:
*True*

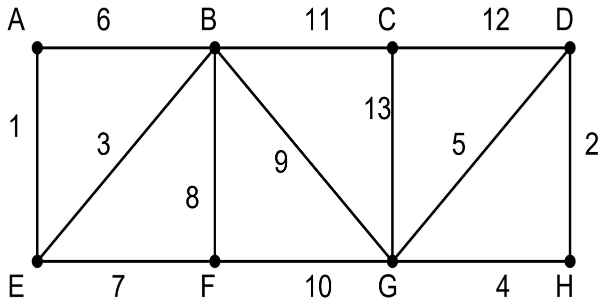
4) The maximum number of union operation one can perform on the partition {1,2},{3,4},{5,6},{7,8},{9,10},{11,12},{13,14},{15,16} is ***1 point***

- 16
- 15
- 8
- 7

Yes, the answer is correct.
Score: 1

Accepted Answers:
*7*

5) Consider the following graph:




The cost of the minimum spanning tree is _____

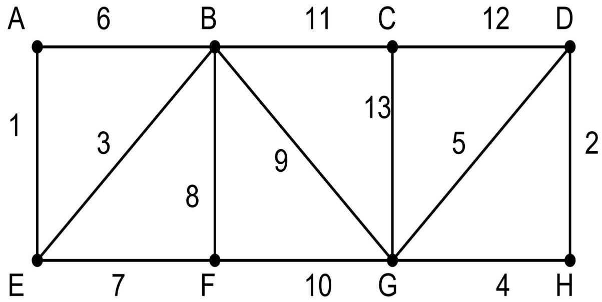
30

No, the answer is incorrect.
Score: 0

Accepted Answers:
*(Type: String) 37*

***1 point***

6) Consider the following graph:




The weight of the second edge ignored by Kruskal's algorithm is

8

No, the answer is incorrect.
Score: 0

Accepted Answers:
*(Type: String) 6*

***1 point***

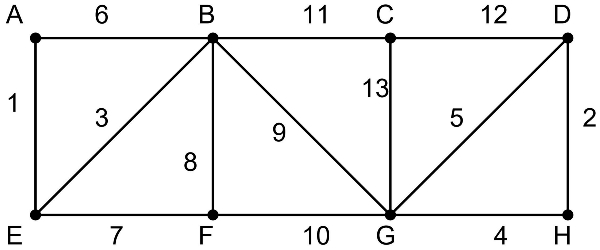
unit=23&lesson=31)

**Week 7 DFS and Cut Vertex ()**

**Video download ()**

**Live session ()**

7) Consider the following graph: ***1 point***



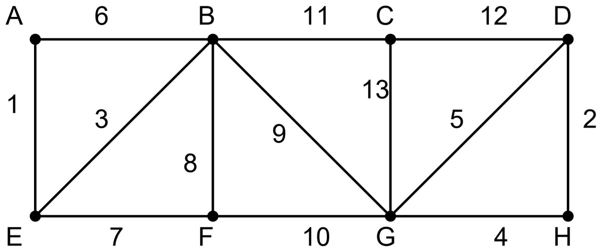

State true or false:
The edge *(B,F)* is in MST.

- False
- True

Yes, the answer is correct.
Score: 1

Accepted Answers:
*False*

8) Consider the following graph: ***1 po***




State true or false:
The edge *(B,G)* is in MST.

- False
- True

Yes, the answer is correct.
Score: 1

Accepted Answers:
*True*

9) Execute the Prim's algorithm on the graph given, starting from *A* and call the spanning tree as $T_1$. Execute the Prim's algorithm on the same graph starting from *D* and call the tree as $T_2$. Which of the following is True? ***2 points***

- $T_1$ has *(A,E)* and $T_2$ does not contain *(A,E)*
- $T_2$ has *(D,H) but* $T_1$ *does not contain (D,H)*
- $T_1$ and $T_2$ are same
- $T_1$ and $T_2$ are different

Yes, the answer is correct.
Score: 2

Accepted Answers:
*$T_1$ and $T_2$ are same*